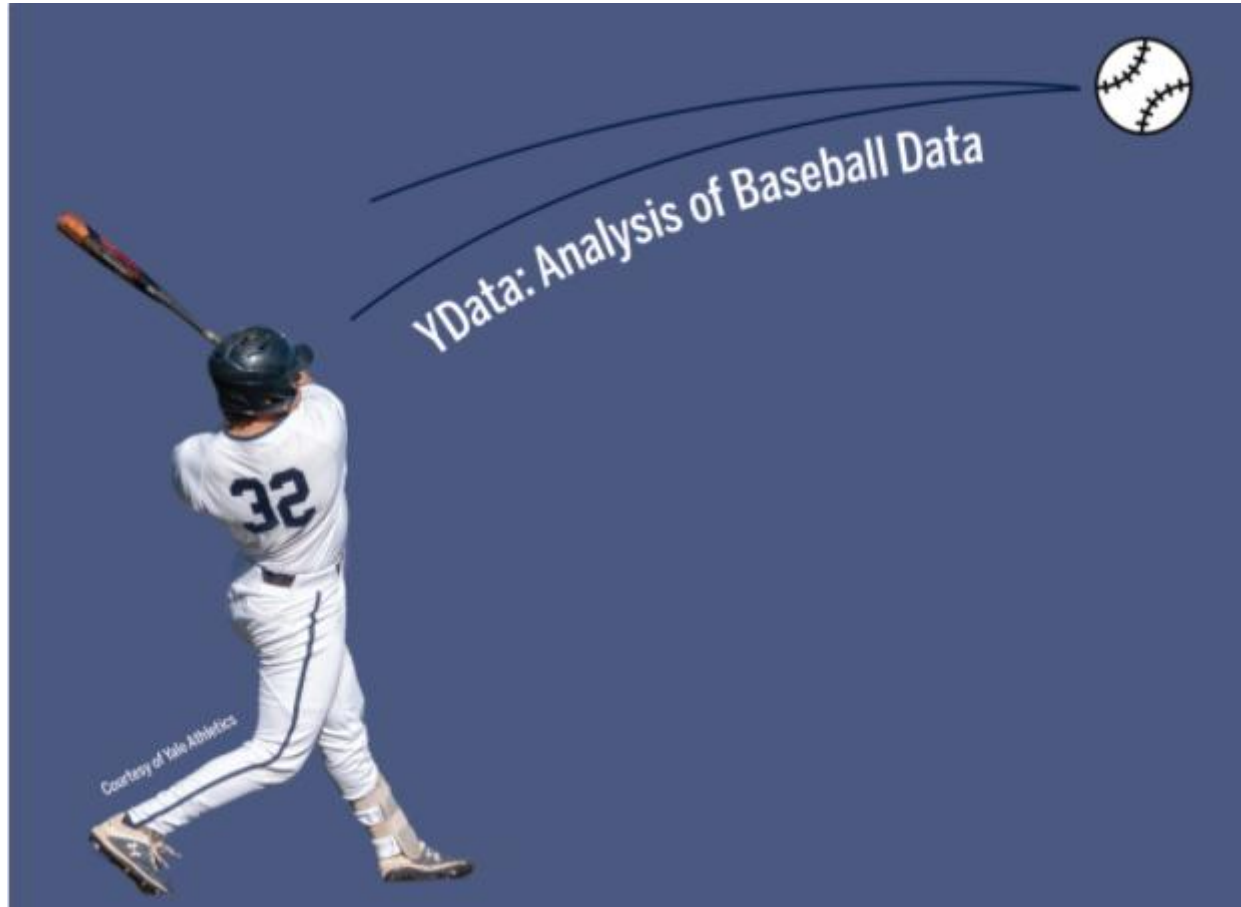


S&DS 1730



Class 6: Probability and simulations using games part I

Overview

Discussion of chapter 4 of Astrobball

Discussion of probability

Big League Baseball

Simulating big league baseball

Object oriented programming

If there is time: work on lab 5

Announcement: Lab 5

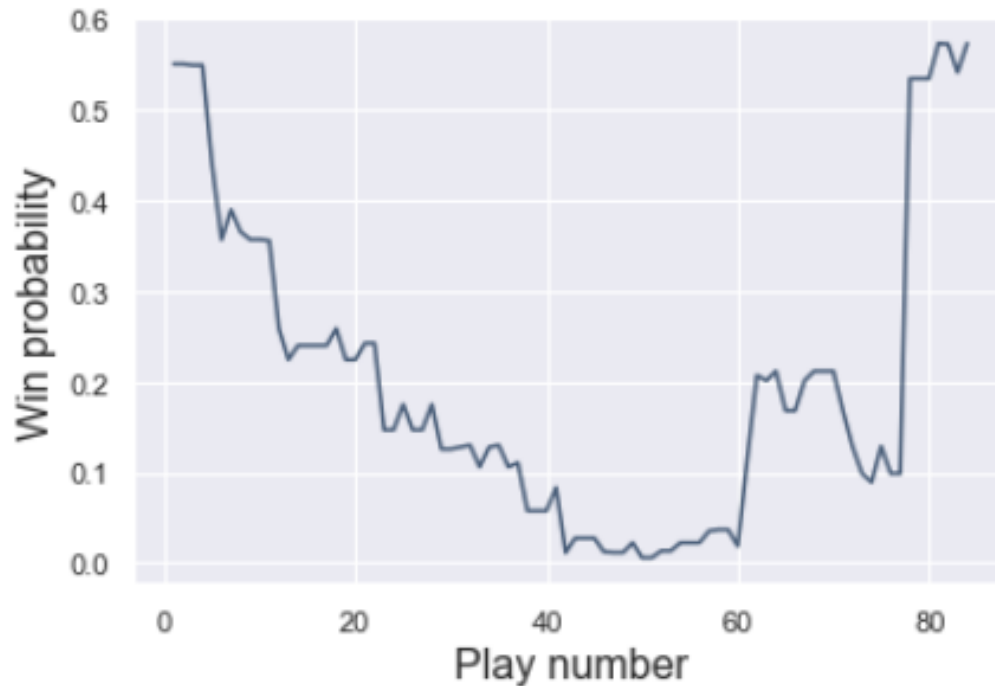
Lab 5 has been released. It is due on
Sunday February 22nd

How is did lab 4 go?

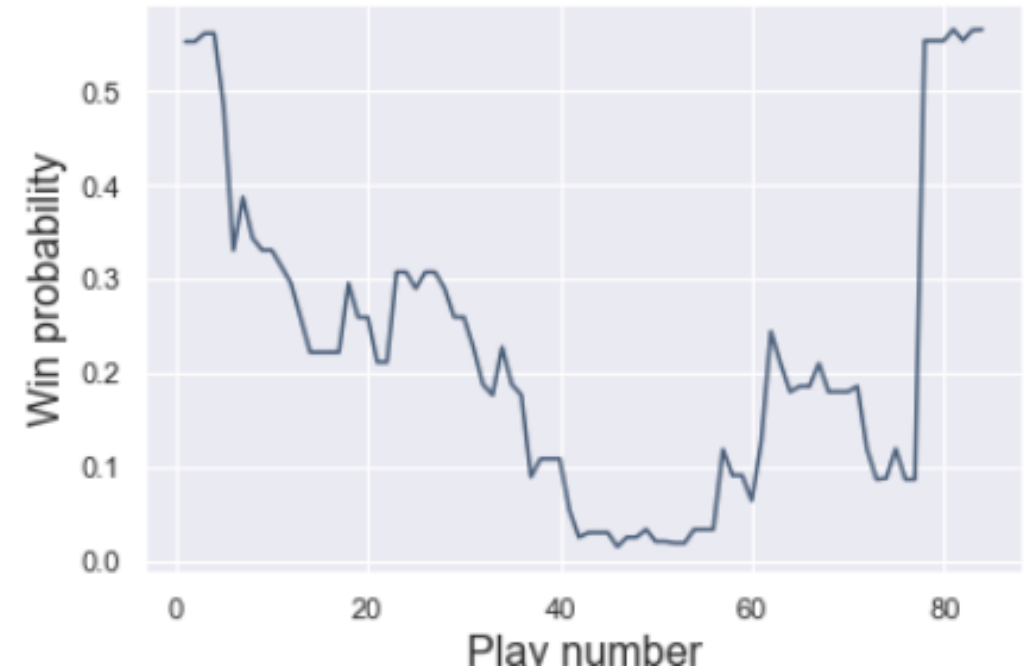


Lab 4: questions?

Using data from 2019
to get probabilities



Using data from 1988
to get probabilities

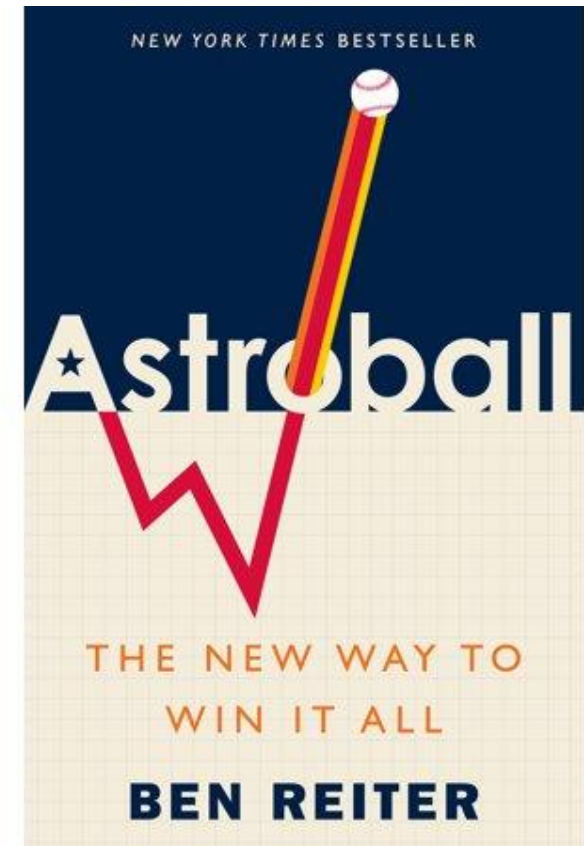


Probability of a come back from down by 6 in the 6th using data from 1988 is: 3.28% vs. 2.2% for 2019

Discussion on chapter 4 of Astrobball

Let's discuss the chapter 4 for 3 minutes in small groups of 3-4 people and then have a larger conversation as a group

- E.g., Discuss your quote and reaction to chapter 4

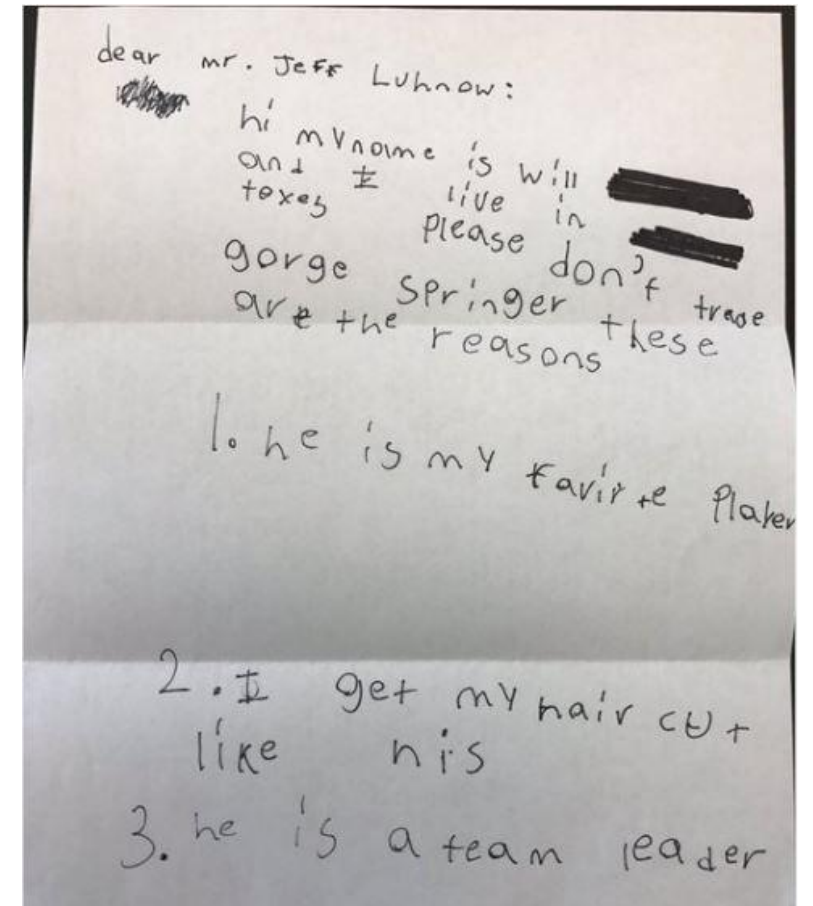


Thoughts on the chapter 4 of Astrobball?

Thoughts on the chapter 4 of Astrobball?

Roster ruthlessness:

"It is purely statistical analysis, " Lowrie told Evan Drellich, the *Houston Chronicle's* beat writer, of the way Luhnow and the Astros operated. "I think you can't have that approach and expect to have good personal relations." pg 90



Thoughts on the chapter 4 of Astroball?

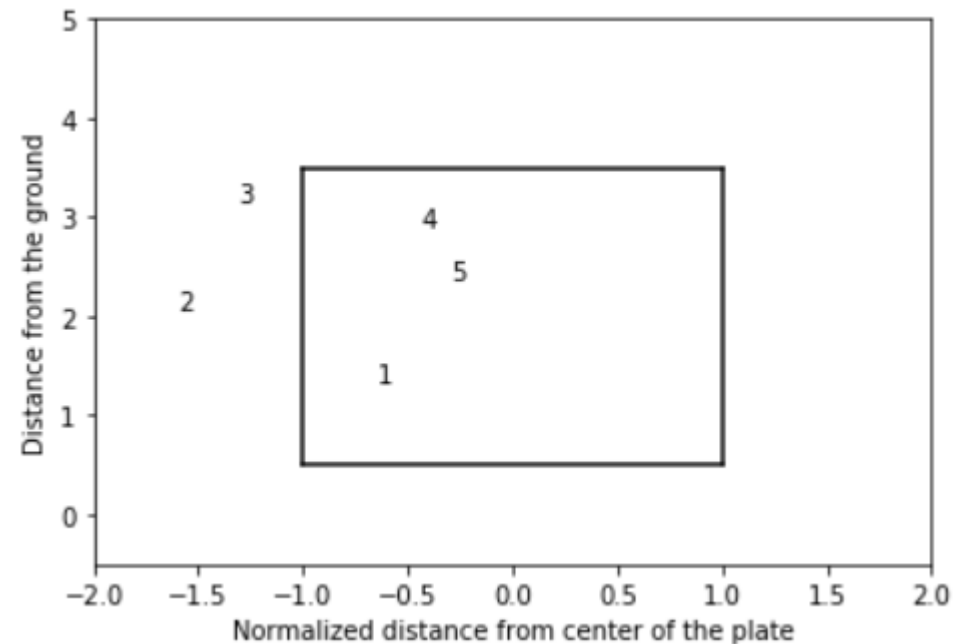
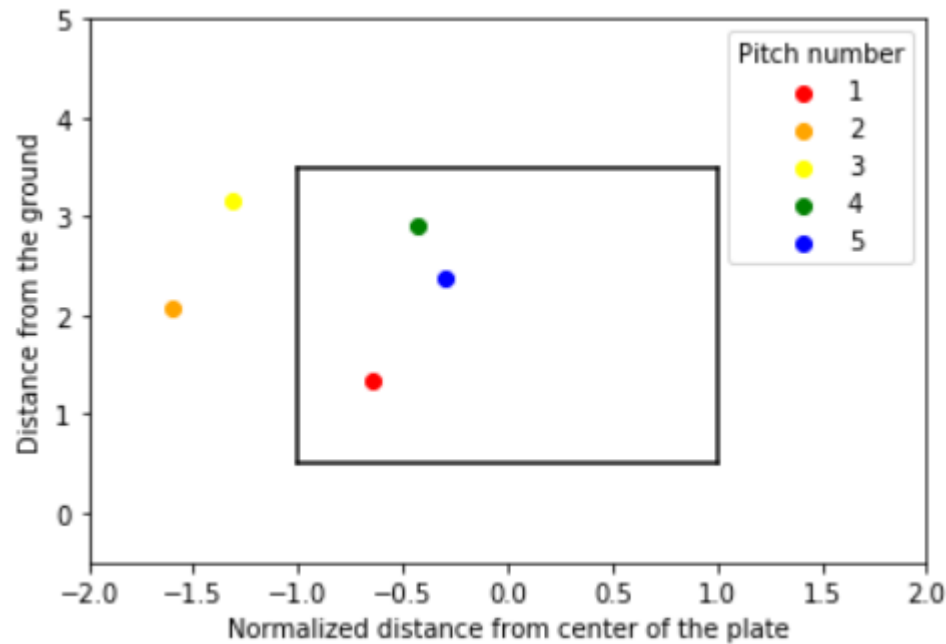
Keuchel and the evolution of baseball video games

- Page 92



Thoughts on the chapter 4 of Astroball?

Keuchel vs. Fielder pitch locations using Statcast data:



Probability

Probability is a way of measuring the likelihood that an event will occur

Probability models can be used to:

- Determine the likelihood of an event
- Simulate events

Definitions:

Sample space: All possible outcomes that can occur

- E.g., all outcomes that could happen as an outcome of a plate appearance

Event: A subset of the sample space

- E.g., A hit is {single, double, triple, home run}



Probability

Probability models assigns a number between 0 and 1 to the outcome of an event occurring:

$P(\text{event}) = 0$ if there is no chance of an event occurring

- E.g., The probability that Aaron Judge will hit 100 home runs in a game

$P(\text{event}) = 1$ if the event will definitely occur

- E.g., The probability that a strike will be thrown in the 2026 baseball season

$P(\text{event}) \in [0, 1]$ if there is some possibility that an event will occur

- E.g. the probability that Aaron Judge will hit a home run on his first plate appearance this season

Probability

One way to interpret probability is in terms of the **relative frequency** of an event.

If we repeat an experiment N times, we can get an estimate of the probability:

$$P(\text{event}) \approx \frac{\text{Number of times the event occurred}}{N}$$

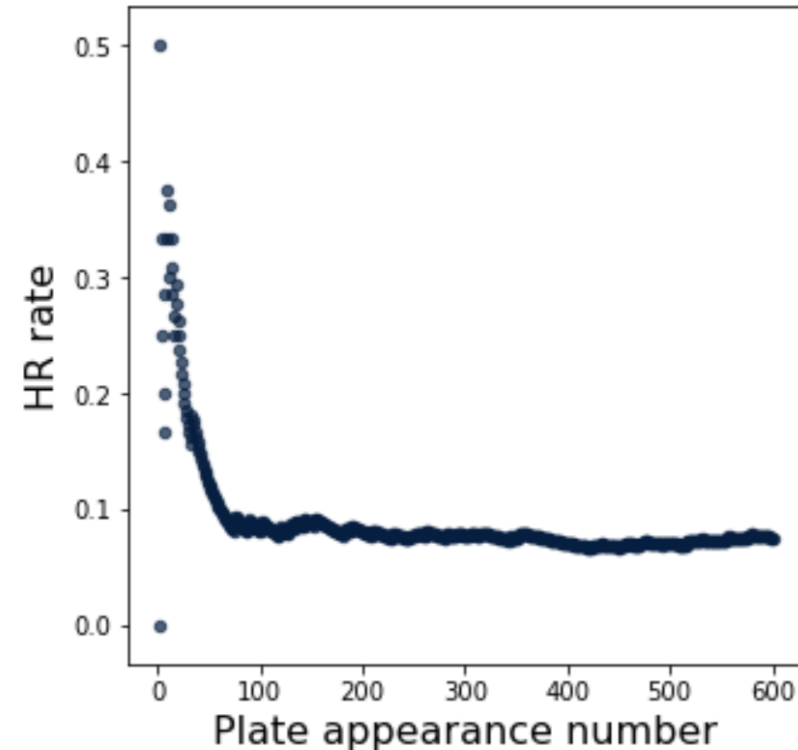
If we repeated this infinitely many times, we would get the true probability of the event

- i.e., $N \rightarrow \infty$

Probability

For example, we can estimate the probability that Mike Trout will hit a home.

$$P(\text{HR}) \approx \frac{\text{Number of HRs}}{\text{Number of PAs}}$$



Random variables

Often, we map the random outcomes into numbers:

- HR: $X = 1$
- Not a home run: $X = 0$

These random numbers are called **random variables**

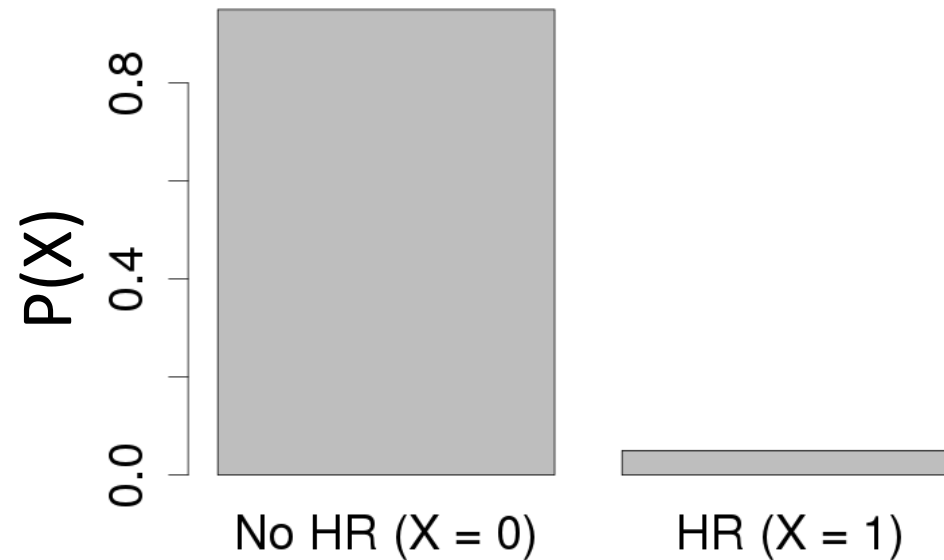
- They are often denoted with capital letters such as X

We then refer to the probability of getting a particular number

- $P(\text{HR}) = P(X = 1) = .10$

Parametric probability distributions

For all probability distributions, the sum of the probability measures over all outcomes (i.e., over the whole sample space) must equal 1



Bernoulli Distribution

Example of rolling a die

Example: If we roll a fair 6-sided die:

What is the sample space?

What are the probability of the outcomes?

Roll outcome k	1	2	3	4	5	6
Probability: $P(X = k)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

Probability rules - Additive rule

If there are two events A, and B, then the probability of A *or* B happening is:

Probability of A	Probability of B	Probability of A <i>and</i> B
		
$P(A \text{ or } B) = P(A) + P(B) - P(A, B)$		

Example 1: what is the probability of rolling a 1 or 2?

- $P(A) = P(\text{rolling } X = 1) = 1/6$
- $P(B) = P(\text{rolling } X = 2) = 1/6$
- $P(A \text{ and } B) = P(\text{rolling } X = 1 \text{ and } X = 2) = 0$

Probability rules - Additive rule

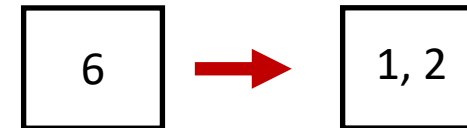
If there are two events A, and B, then the probability of A **or** B happening is:

$$\begin{array}{ccccc} \text{Probability of A} & & \text{Probability of B} & & \text{Probability of A \textit{and} B} \\ & \searrow & \searrow & & \searrow \\ P(A \text{ or } B) & = & P(A) & + & P(B) & - & P(A, B) \end{array}$$

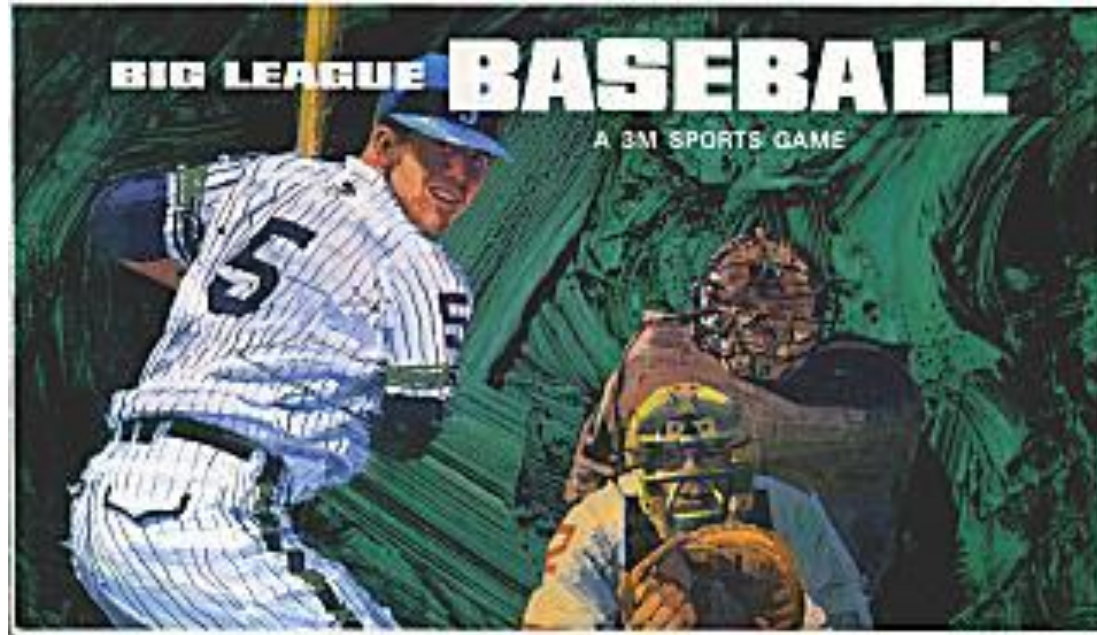
Example 2:

- Suppose we replace the 6 on a die with the number 1 and 2
- What is the probability of getting a 1 or a 2?

$$\begin{aligned} &P(X = 1) + P(X = 2) - P(X = 1 \text{ and } 2) \\ &2/6 + 2/6 - 1/6 = 3/6 = 1/2 \end{aligned}$$



Big League Baseball



Big league baseball was a board game from the 1960's

Events in a baseball game were modeled as the outcome of rolling dice

Big League Baseball

Rules for Big League Baseball

A single die is rolled:

- If a 2 or 3 occurs: a ball is pitched
- If a 4 or 5 occurs: a strike is pitched
- If a 1 or 6 occurs: a ball is hit in fair play

What is the sample space?

What is the probability of a ball being hit in fair play?



Big League Baseball

Rules for Big League Baseball

If a fair ball is hit (1 or 6 rolled first) then two dice are rolled in sequence and the following table indicates the outcome of the play:

		2nd Die					
1st Die		1	2	3	4	5	6
	1	Single	Out	Out	Out	Out	Error
	2	Out	Double	Single	Out	Single	Out
	3	Out	Single	Triple	Out	Out	Out
	4	Out	Out	Out	Out	Out	Out
	5	Out	Single	Out	Out	Out	Single
	6	Error	Out	Out	Out	Single	Home run

Big League Baseball

If a fair ball is hit, what is the probability of the following events:

- A home run?
- A out?
- A single?
- A hit?

		2nd Die					
1st Die		1	2	3	4	5	6
	1	Single	Out	Out	Out	Out	Error
	2	Out	Double	Single	Out	Single	Out
	3	Out	Single	Triple	Out	Out	Out
	4	Out	Out	Out	Out	Out	Out
	5	Out	Single	Out	Out	Out	Single
	6	Error	Out	Out	Out	Single	Home run

Big League Baseball

If a fair ball is hit, what is the probability of the following events:

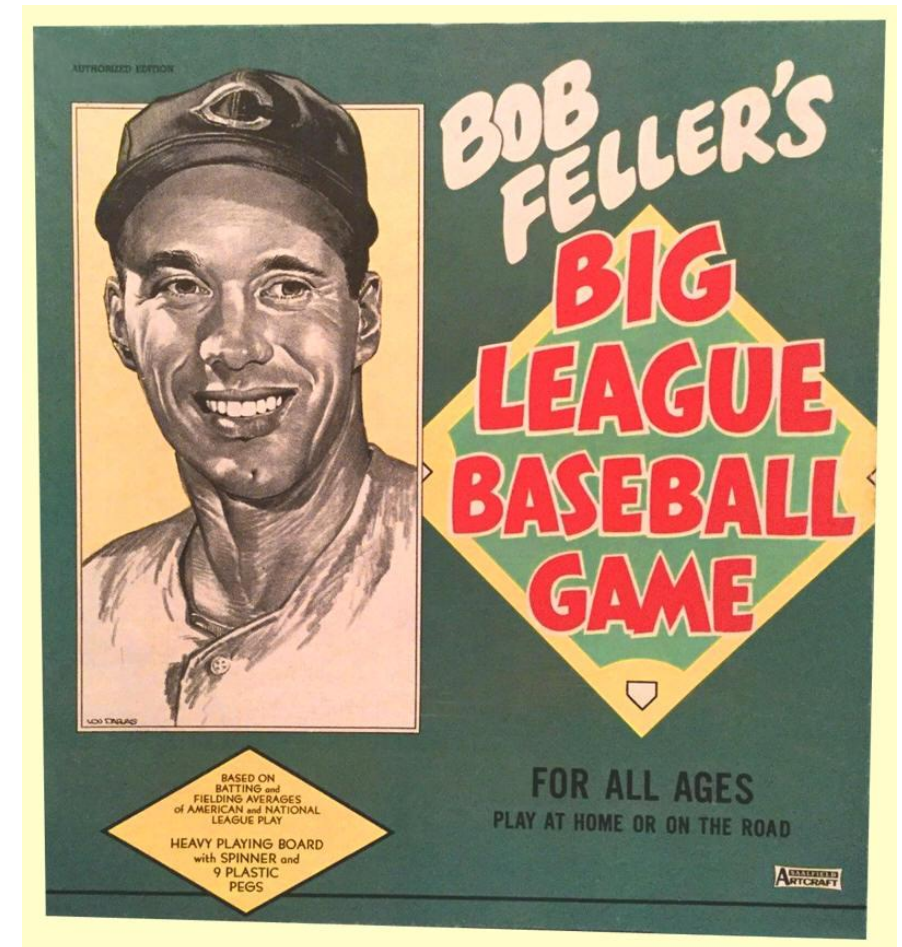
- A home run? $1/36$
- A out? $24/36$
- A single? $7/36$
- A hit? $10/36$

		2nd Die					
1st Die		1	2	3	4	5	6
	1	Single	Out	Out	Out	Out	Error
	2	Out	Double	Single	Out	Single	Out
	3	Out	Single	Triple	Out	Out	Out
	4	Out	Out	Out	Out	Out	Out
	5	Out	Single	Out	Out	Out	Single
	6	Error	Out	Out	Out	Single	Home run

Big League Baseball

Let's play an inning of Big League Baseball

- Keep score with the scorecard!
- Rules of Big League Baseball and a pdf of a scorecard are also on Canvas



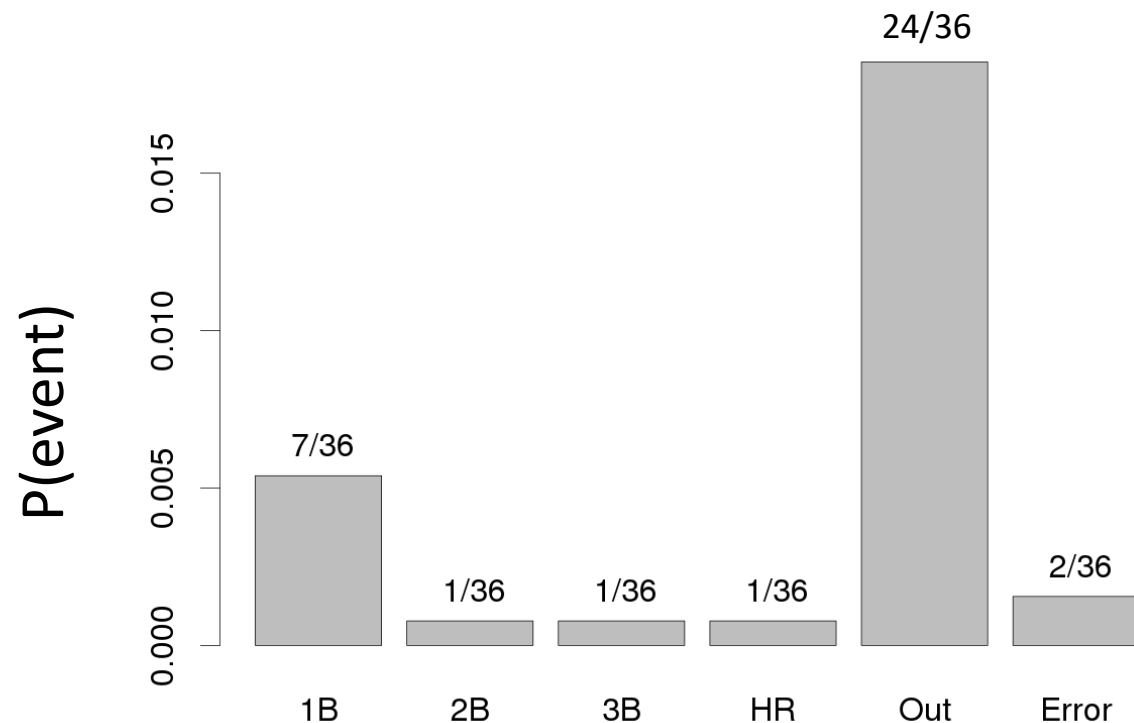
Limitations of Big League Baseball

What are a few limitations of Big League Baseball?

- All batters have the same ability
- All pitchers have the same ability
- Balls and strikes are equally likely
 - In real baseball many more strikes are thrown



Big League baseball probability distribution if a ball is hit in play



Q: why is this a valid probability distribution?

Big League Baseball

Suppose we want to calculate the probability of a home run, *without assuming that a ball was hit in play*

What are the sequences of events that would lead to a home run?

1) We need to hit a ball in play (roll a 1 or a 6)

- $P(\text{ball in play}) = 1/3$

2) We need to hit a home run (roll two 6's)

- $P(\text{home run} \mid \text{ball in play}) = 1/36$

We somehow need to combine these events to get a total probability

Multiplicative Rule

The probability of two events A *and* B occurring is:

- The probability of B occurring ... times ...
- The probability of A occurring given that B has occurred

$$P(A, B) = P(A|B) \times P(B)$$

E.g., Suppose we draw 2 cards from a 52 card deck. What is the probability they are both diamonds?

$$\begin{aligned} P(D_1, D_2) &= P(D_1) \times P(D_2 | D_1) \\ &= 13/52 \times 12/51 = 0.12 \end{aligned}$$

Special Case: Multiplicative Rule for independent events

Two events are called **independent** if:

$$P(A, B) = P(A) \times P(B)$$

Q: what is the probability of getting two strikes in a row in Big League baseball at the start of a plate appearance?

$$\begin{aligned} P(S_1, S_2) &= P(S_1) \times P(S_2) \\ &= 1/3 \times 1/3 = 1/9 \end{aligned}$$

Big League Baseball

What is the probability one would strike out on three straight pitches?

- Answer: $(1/3)^3 = 1/27$

At the start of a plate appearance, what is the probability one would strike out on 4 pitches?

- Probability of a strike is $1/3$
- Probability of a ball is $1/3$
- Number of ways to strike out on 4 pitches is:
 - B,S,S,S or
 - S,B,S,S or
 - S,S,B,S
- Probability is: $(1/3)^4 + (1/3)^4 + (1/3)^4 = 1/27$

Simulating Big League Baseball



Simulating Big League Baseball

To simulate Big League baseball we will need to:

1. Simulate rolls of the dice in Python:

```
import numpy as np
```

```
np.random.choice(np.arange(1, 7))
```


Simulating Big League Baseball

2. Use conditional statements:

```
if (X == 1) or (Y == 2):  
    # do stuff
```

```
elif (X == 3) or (Y == 4):  
    # do different stuff
```

```
else:  
    # do this stuff
```

3. Use dictionaries to map die rolls to the names of outcomes

```
play_dictionary = { (1, 1) : "single",  
                    (1, 6) : "error"  
                    }
```

You will try it on lab 5!

Object oriented programming



Keeping track of the state of a game

If we want to do a full simulation of a Big League Baseball game it will be useful to keep track of the game after every event

We can use Object-oriented programming (OOP) to do this!

Object-oriented programming

Object-oriented programming (OOP) is a programming paradigm based on the concept of "objects", which can contain:

1. Data: in the form of fields
 - often known as attributes or properties
2. Methods: functions that operate on the data

Have we seen any objects in this class yet?

[illegible]

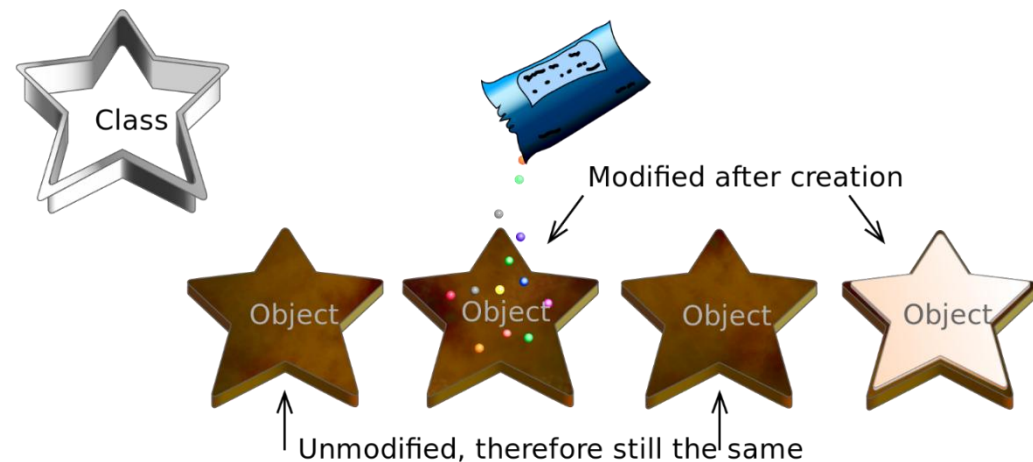
Object-oriented programming

A **class** defines what the object is

- i.e., classes have code that lists the types of data stored and the methods

An **object** is a realization of classes (often called an instance of a class)

- Stores actual data that can be manipulated with methods
- E.g., `my_df.sum()`



Object-oriented programming

A **constructor** defines what should happen when an object is created

- In Python the constructor is defined with the `__init__(self)` method

```
class Baseball_Game:
```

```
    # Constructor
```

```
    def __init__(self):
```

```
        self.end_of_game = False
```

```
        self.inning = 1
```

```
        ...
```

```
# create an instance of an object
```

```
my_object = Baseball_Game()
```

Object-oriented programming

The `__str__` method defines what should happen when the `print()` function is called on the object

```
# str method
def __str__(self):
    'I am an object'
```

```
# create an instance of an object
my_object = Baseball_Game()

# print the object
print(my_object)
```

Object-oriented programming

Let's begin on **Lab 5 Exercise 3** where you will create an object to keep track of the state of the baseball game